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Solid-state imaging device, method for driving solid-state imaging device, and electronic apparatus

Abstract

A pixel circuit **200** includes a readable pixel **210**, a comparator **220**, and a selector counter circuit **230**. The readable pixel **210** performs photoelectric conversion at a photodiode PD**11** and produces a readable signal corresponding to an illuminance condition of incident light. The readable pixel **210** includes an overflow path extending to a floating diffusion FD**11**. The comparator **220** compares a voltage signal (SFout) read out from the readable pixel **210** against a reference signal Vref and outputs a comparison result signal Vout indicating the result of the comparison. The selector counter circuit **230** includes a selector circuit for selecting an external clock or the output Vout from the comparator and a counter circuit for counting the output from the selector circuit.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS
(1) This application is the U.S. National Phase Application of PCT/JP2022/017040, filed on Apr. 4, 2022, which claims the benefit of priority from Japanese Patent Application Serial No. 2021-064260 (filed on Apr. 5, 2021), the contents of which are incorporated herein by reference in entirety.
TECHNICAL FIELD
(2) The present invention relates to a solid-state imaging device, a method for driving a solid-state imaging device, and an electronic apparatus.
BACKGROUND
(3) Solid-state imaging devices (image sensors) including photoelectric conversion elements for detecting light and generating charges are embodied as complementary metal oxide semiconductor (CMOS) image sensors, which have been in practical use. The CMOS image sensors have been widely applied as parts of various types of electronic apparatuses such as digital cameras, video cameras, surveillance cameras, medical endoscopes, personal computers (PCs), mobile phones and other portable terminals (mobile devices).
(4) The CMOS image sensors include, for each pixel, a photodiode (a photoelectric conversion element) and a floating diffusion (FD) amplifier having a floating diffusion (FD). The mainstream design of the reading operation in the CMOS image sensors is a column parallel output processing of selecting one of the rows in the pixel array and reading the pixels in the selected row simultaneously in the column output direction.
(5) Various types of pixel signal reading (output) circuits have been proposed for CMOS image sensors of the column parallel output scheme. Among them, one of the most advanced circuits is a circuit that includes an analog-to-digital converter (ADC) for each column and obtains a pixel signal in a digital format (see, for example, Patent Literatures 1 and 2).
(6) In this CMOS image sensor having column-parallel ADCs (column-wise-AD CMOS image sensor), AD conversion is performed in such a manner that a comparator compares the pixel signal

(voltage signal) against a so-called RAMP wave and a counter of a later stage performs digital CDS.
(7) The CMOS image sensors of this type are capable of transferring signals at high speed, but disadvantageously incapable of reading the signals with a global shutter.
(8) To address this issue, a digital pixel sensor has been proposed that has, in each pixel, an ADC including a comparator (and additionally a memory part), so that the sensor can realize a global shutter according to which the exposure to light can start and end at the same timing in all of the pixels of the pixel array part (see, for example, Patent Literatures 3 and 4).
(9) The above-described conventional CMOS image sensor constituted by a digital pixel sensor is capable of realizing global shutter function. In addition, if an ADC including a comparator is arranged in each pixel and reading is performed in a predetermined mode, the conventional CMOS image sensor is capable of achieving widened dynamic range.
(10) The dynamic range can be widened by, for example, reading two types of signals having different integration durations from the same pixel of the image sensor and combining the read two types of signals, or by combining a signal with a small dynamic range read from a high-sensitive pixel and a signal with a widened dynamic range read from a low-sensitive pixel.
(11) Patent Literature 5 presents a unit pixel configuration for capturing infrared ray images. The unit pixel includes two input stages covering both low and high light levels, and also an internal automatic input selection circuit for increasing the dynamic range. Patent Literature 5 discloses, as feasible digital implementation, a technique of applying pulse frequency modulation (PFM) ADC for the unit pixel.
(12) Patent Literature 6 discloses a digital read-out system pixel circuit including an extension counter and PFM capability for accomplishing excellent quantization noise characteristics.
(13) FIGS. **1** and **2** are here referred to in order to describe the basic configuration and function of the PFM. FIG. **1** shows the configuration of a digital read-out system pixel circuit having a PFM function according to a first example. FIG. **2** shows the configuration of a digital read-out system pixel circuit having a PFM function according to a second example.
(14) A pixel circuit **1** shown in FIG. **1** includes a photodiode PD**1**, a reset transistor RST-Tr, a comparator **2**, and a counter **3**.
(15) In the pixel circuit **1**, the comparator **2** compares the voltage at the charge storage node ND**1** of the photodiode PD**1** against reference voltage Vref**1** (=VSA**1**), which is equivalent to the full well voltage VSA**1** of the photodiode PD**1**. If the node voltage ND**1** reaches the full well voltage VSA**1**, the output from the comparator **2** is fed back to the gate of the reset transistor RST-Tr as a feedback reset signal FRST, which brings the reset transistor RST-Tr into the conduction state. Stated differently, the feedback resetting can reset the charges stored in the photodiode PD**1**. The counter **3** counts how many times the photodiode PD**1** is reset.
(16) A pixel circuit **1**A shown in FIG. **2** has a transfer transistor TG-Tr connected between the charge storage node ND**1** of the photodiode PD**1** and a floating diffusion FD**1** serving as an output node. A control signal TG can control whether the transfer transistor TG-Tr is in the conduction state. In the pixel circuit **1**A, a logic circuit **4** is provided in the feedback channel of the feedback reset signal FRST, and delay circuits **5-1** and **5-2** are provided and connected to the output side of the comparator **2** and to the input side of the logic circuit **4** to address the delays caused by the logic circuit **4**.
(17) In the pixel circuit **1**A, the transfer transistor TG-Tr is controlled to remain in the conduction state (on state) during the exposure period and in the non-conduction state (off state) during the reset phase of the floating diffusion FD**1**.
RELEVANT REFERENCE
List of Relevant Patent Literature
(18) Patent Literature 1: Japanese Patent Application Publication No. 2005-278135 Patent Literature 2: Japanese Patent Application Publication No. 2005-295346 Patent Literature 3: U.S.

Pat. No. 7,164,114 B2 FIG. 4 Patent Literature 4: US 2010/0181464 A1 Patent Literature 5: U.S. Pat. No. 9,324,745 B2 Patent Literature 6: U.S. Pat. No. 9,197,834 B2

SUMMARY

(19) The pixel circuits **1** and **1**A shown in FIGS. **1** and **2**, however, present the following disadvantages.

(20) As configured without a transfer transistor TG-Tr, the pixel circuit **1** shown in FIG. **1** is significantly disturbed by the dark current of the photodiode PD**1**. There are difficulties in reading low signals produced by the photodiode PD**1** under a low illuminance condition in the above-described manner.

(21) Although the pixel circuit **1**A shown in FIG. **2** has a transfer transistor TG-Tr, the dark current of the photodiode PD**1** at the silicon interface under the transfer transistor TG-Tr is still a critical issue. Likewise, there are difficulties in reading low signals produced by the photodiode PD**1** under a low illuminance condition in the above-described manner. As including the logic circuit (NAND, NOR) **4** for controlling the transfer transistor TG-Tr, the pixel circuit **1**A faces serious difficulties in achieving a reduced pixel size.

(22) The present invention is designed to provide a solid-state imaging device, a method for driving a solid-state imaging device, and an electronic apparatus that are capable of reading low signals produced under a low illuminance condition with low noise while keeping the sensitivity and full well unchanged and also capable of achieving a reduced pixel size.

(23) A first aspect of the present invention provides a solid-state imaging device including: a readable pixel configured to perform photoelectric conversion, the readable pixel being configured to produce a readable signal corresponding to an illuminance condition of incident light; a comparator configured to compare a voltage signal read from the readable pixel against a reference signal and output a comparison result signal corresponding to a result of the comparison; and a selector counter circuit including a selector circuit and a counter circuit, the selector circuit being configured to select an external clock or an output from the comparator, the counter circuit being configured to count an output from the selector circuit. The readable pixel includes: a photoelectric conversion element configured to store therein, in an exposure period, charges corresponding to an amount of the incident light; a transfer element configured to remain in a non-conduction state during the exposure period and to remain in a conduction state during a transfer period to transfer the charges stored in the photoelectric conversion element; a floating diffusion configured to hold the charges transferred thereto via the transfer element so that the charges are read out as a voltage signal; at least one reset element configured to perform a reset operation of discharging the charges stored in the floating diffusion; and an overflow path configured to allow charges overflowing from the photoelectric conversion element to overflow toward a region where the floating diffusion is formed. The readable pixel, the comparator, the selector circuit and the counter circuit: under a high illuminance condition, if the voltage signal corresponding to the charges held in the floating diffusion reaches reference voltage, perform a pulse frequency modulation (PFM) mode operation, according to which the comparison result signal from the comparator is applied to perform feedback resetting to allow the floating diffusion to perform self-resetting, and the selector counter circuit counts a reset frequency of the floating diffusion in response to the comparison result signal; and under a low illuminance condition, perform a dual sampling read-out mode operation using the charges stored in the photoelectric conversion element and overflow charges.

(24) A second aspect of the present invention provides a method for driving a solid-state imaging device. The solid-state imaging device includes: a readable pixel configured to perform photoelectric conversion, the readable pixel being configured to produce a readable signal corresponding to an illuminance condition of incident light; a comparator configured to compare a voltage signal read from the readable pixel against a reference signal and output a comparison result signal corresponding to a result of the comparison; and a selector counter circuit including a selector circuit and a counter circuit, the selector circuit being configured to select an external clock

or an output from the comparator, the counter circuit being configured to count an output from the selector circuit. The readable pixel includes: a photoelectric conversion element configured to store therein, in an exposure period, charges corresponding to an amount of the incident light; a transfer element configured to remain in a non-conduction state during the exposure period and to remain in a conduction state during a transfer period to transfer the charges stored in the photoelectric conversion element; a floating diffusion configured to hold the charges transferred thereto via the transfer element so that the charges are read out as a voltage signal; at least one reset element configured to perform a reset operation of discharging the charges stored in the floating diffusion; and an overflow path configured to allow charges overflowing from the photoelectric conversion element to overflow toward a region where the floating diffusion is formed. Under a high illuminance condition, if the voltage signal corresponding to the charges held in the floating diffusion reaches reference voltage, a pulse frequency modulation (PFM) mode operation is performed, according to which the comparison result signal from the comparator is applied to perform feedback resetting to allow the floating diffusion to perform self-resetting, and the selector counter circuit counts a reset frequency of the floating diffusion in response to the comparison result signal. Under a low illuminance condition, a dual sampling read-out mode operation is performed using the charges stored in the photoelectric conversion element and overflow charges.
(25) A third aspect of the present invention provides an electronic apparatus including: a solid-state imaging device; and an optical system for forming a subject image on the solid-state imaging device. The solid-state imaging device includes: a readable pixel configured to perform photoelectric conversion, the readable pixel being configured to produce a readable signal corresponding to an illuminance condition of incident light; a comparator configured to compare a voltage signal read from the readable pixel against a reference signal and output a comparison result signal corresponding to a result of the comparison; and a selector counter circuit including a selector circuit and a counter circuit, the selector circuit being configured to select an external clock or an output from the comparator, the counter circuit being configured to count an output from the selector circuit. The readable pixel includes: a photoelectric conversion element configured to store therein, in an exposure period, charges corresponding to an amount of the incident light; a transfer element configured to remain in a non-conduction state during the exposure period and to remain in a conduction state during a transfer period to transfer the charges stored in the photoelectric conversion element; a floating diffusion configured to hold the charges transferred thereto via the transfer element so that the charges are read out as a voltage signal; at least one reset element configured to perform a reset operation of discharging the charges stored in the floating diffusion; and an overflow path configured to allow charges overflowing from the photoelectric conversion element to overflow toward a region where the floating diffusion is formed. The readable pixel, the comparator, the selector circuit and the counter circuit: under a high illuminance condition, if the voltage signal corresponding to the charges held in the floating diffusion reaches reference voltage, perform a pulse frequency modulation (PFM) mode operation, according to which the comparison result signal from the comparator is applied to perform feedback resetting to allow the floating diffusion to perform self-resetting, and the selector counter circuit counts a reset frequency of the floating diffusion in response to the comparison result signal; and under a low illuminance condition, perform a dual sampling read-out mode operation using the charges stored in the photoelectric conversion element and overflow charges.
ADVANTAGEOUS EFFECTS
(26) According to the present invention, low signals produced under a low illuminance condition can be read with low noise while the sensitivity and full well are kept unchanged and a reduced pixel size can be also achieved.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. **1** shows the configuration of a digital read-out system pixel circuit having a PFM function according to a first example.

(2) FIG. **2** shows the configuration of a digital read-out system pixel circuit having a PFM function according to a second example.

(3) FIG. **3** is a block diagram showing an example configuration of a solid-state imaging device according to a first embodiment of the present invention.

(4) FIG. **4** is a block diagram showing an example basic configuration of a pixel circuit of the solid-state imaging device according to the first embodiment of the present invention.

(5) FIG. **5** is a circuit diagram showing an example configuration of a readable pixel of the pixel circuit relating to the first embodiment of the present invention.

(6) FIG. **6** is a circuit diagram showing an example configuration of a comparator relating to the first embodiment of the present invention.

(7) FIG. **7** is a simplified sectional view showing an example configuration of the readable pixel according to the first embodiment of the present invention.

(8) FIG. **8** is a timing chart to illustrate an example sequence of operations for reading performed in a PFM mode and a dual sampling read-out mode on the pixel circuit under a high illuminance condition in the solid-state imaging device relating to the first embodiment of the present invention.

(9) FIG. **9** illustrates a sequence of operations and potential transition to explain the operations performed in the PFM mode and dual sampling read-out mode under a high illuminance condition on the pixel circuit of the solid-state imaging device relating to the first embodiment of the present invention.

(10) FIG. **10** is a timing chart to illustrate an example sequence of operations for reading performed in the PFM mode and dual sampling read-out mode on the pixel circuit under a low illuminance condition in the solid-state imaging device relating to the first embodiment of the present invention.

(11) FIG. **11** illustrates a sequence of operations and potential transition to explain the operations performed in the PFM mode and dual sampling read-out mode under a low illuminance condition on the pixel circuit of the solid-state imaging device relating to the first embodiment of the present invention.

(12) FIG. **12** shows linearity exhibited by a combination signal of read-out signals in the dual sampling read-out mode in the solid-state imaging device relating to the first embodiment of the present invention.

(13) FIG. **13** is a circuit diagram showing an example configuration of a readable pixel of a pixel circuit relating to a second embodiment of the present invention.

(14) FIG. **14** is a simplified sectional view showing an example configuration of a charge integrating and transferring system including a shutter gate transistor, or the main part of the pixel circuit relating to the second embodiment of the present invention.

(15) FIG. **15** is a timing chart to illustrate an example sequence of operations for reading performed in the PFM mode and dual sampling read-out mode on the pixel circuit under a high luminance condition in the solid-state imaging device relating to the second embodiment of the present invention.

(16) FIG. **16** is a timing chart to illustrate an example sequence of operations for reading performed in the PFM mode and dual sampling read-out mode on the pixel circuit under a low illuminance condition in the solid-state imaging device relating to the second embodiment of the present invention.

(17) FIG. **17** is a circuit diagram showing an example configuration of a pixel circuit according to a third embodiment of the present invention.

(18) FIG. **18** is a circuit diagram showing an example configuration of a pixel circuit of a solid-state imaging device relating to a fourth embodiment of the present invention.

(19) FIG. **19** is a circuit diagram showing an example configuration of a pixel circuit of a solid-state imaging device relating to a fifth embodiment of the present invention.
(20) FIG. **20** illustrates an example of a device to which the solid-state imaging devices relating to the embodiments of the present invention are applied.
(21) FIG. **21** is a timing chart to schematically show how the applied device shown in FIG. **20** works.
(22) FIG. **22** shows an example configuration of an electronic apparatus to which the solid-state imaging devices relating to the embodiments of the present invention can be applied.

DESCRIPTION OF THE PREFERRED EMBODIMENTS

(23) Embodiments of the present invention will be hereinafter described with reference to the drawings.

First Embodiment

(24) FIG. **3** is a block diagram showing an example configuration of a solid-state imaging device relating to a first embodiment of the present invention. FIG. **4** is a block diagram showing an example basic configuration of a pixel circuit of the solid-state imaging device according to the first embodiment of the present invention. In this embodiment, a solid-state imaging device **10** is constituted by, for example, a CMOS image sensor.
(25) As shown in FIG. **3**, the solid-state imaging device **10** is constituted mainly by a pixel part **20** serving as an image capturing part, a vertical scanning circuit (a row scanning circuit) **30**, a reading circuit (a column reading circuit) **40**, a horizontal scanning circuit (a column scanning circuit) **50**, and a timing control circuit **60**. Among these components, for example, the vertical scanning circuit **30**, the reading circuit **40**, the horizontal scanning circuit **50**, and the timing control circuit **60** constitute a reading part **70** for reading out pixel signals.
(26) In the solid-state imaging device **10** relating to the first embodiment, the pixel part **20** has pixel circuits **200** arranged therein in a matrix pattern. As will be described in detail, each pixel circuit **200** includes a readable pixel **210**, a comparator **220**, a selector counter circuit **230**, and a memory circuit **240**. The readable pixel **210** performs photoelectric conversion at a photodiode PD**11** serving as a photoelectric conversion element and produces a readable signal corresponding to an illuminance condition of incident light. The readable pixel **210** includes an overflow path extending to a floating diffusion FD **11**. The comparator **220** compares a voltage signal (SFout) read out from the readable pixel **210** against a reference signal Vref and outputs a comparison result signal Vout indicating the result of the comparison. The selector counter circuit **230** selects a counting clock of a counter in response to the comparison result signal Vout from the comparator **220** and performs counting in synchronization with the selected clock. The memory circuit **240** stores digital data produced by the selector counter circuit **230** based on the comparison result signal Vout from the comparator **220**.
(27) The readable pixel **210**, comparator **220** and selector counter circuit **230** operate in a pulse frequency modulation (PFM) mode under a high illuminance condition and in a dual sampling read-out (for example, LOFIC) mode under a low illuminance condition using the charges stored in the photoelectric conversion element and the overflow charges.
(28) In the PFM mode under the high illuminance condition, if the voltage signal determined by the voltage held in the floating diffusion FD**11** reaches the reference voltage of the comparator, the comparison result signal Vout from the comparator **220** is applied to perform feedback resetting FRST**11** to allow the floating diffusion FD**11** to perform self-resetting, and the number of times the floating diffusion FD**11** is reset (the reset frequency of the floating diffusion FD**11**) is counted by the selector counter circuit **230** in response to the comparison result signal Vout from the comparator **220**.
(29) In the solid-state imaging device **10** relating to the first embodiment of the present invention, the pixel circuit **200** has a lateral overflow integration capacitor (LOFIC), which will be described in detail below. Under a low illuminance condition, the pixel circuit **200** operates in the dual

sampling read-out mode (LOFIC mode) using a dual conversion scheme based on the charges stored in the photodiode PD**11** serving as the photoelectric conversion element and the overflow charges.

(30) In the pixel circuit **200**, the readable pixel **210** and selector counter circuit **230** can selectively operate in the PFM mode or dual sampling read-out mode in response to a mode selection signal SMS.

(31) In the PFM mode, a first comparison result signal Vout**11** is applied as a feedback reset signal FRST**11** to bring the reset element into the conduction state, so that the floating diffusion FD**11** is reset to a predetermined potential VAAPIX in the readable pixel **210**.

(32) In the PFM mode, a first reference signal (reference voltage) Vref**11**, which is at a certain (fixed) level and equivalent to the full well voltage of the photodiode PD**11** serving as the photoelectric conversion element, is fed to the comparator **220**. If the voltage signal determined by the charges held in the floating diffusion FD**11** reaches the first reference signal level Vref**11**, the first comparison result signal Vout**11** is output from the comparator **220** to the readable pixel **210** and selector counter circuit **230**. In the dual sampling read-out mode, a second reference signal Vref**12**, which has a ramp waveform and shows a continuous change, is fed to the comparator **220**. The voltage signal determined by the voltage held in the floating diffusion FD**11** is compared against the second reference signal Vref**12**, and the result is output as a second comparison result signal Vout**12** to the selector counter circuit **230**.

(33) The selector counter circuit **230** counts the first comparison result signal Vout**11** from the comparator **220** in the PFM mode, and latches the second comparison result signal Vout**12** in synchronization with a clock having a predetermined frequency in the dual sampling read-out mode.

(34) According to the example shown in FIG. **4**, the selector counter circuit **230** includes a selector **231** and a counter **232**. The selector **231** has an input terminal A connected to the output line of the comparison result signal Vout from the comparator **220**, an input terminal B connected to the feeding line of the clock pulse CP, and an output terminal C connected to the input terminal of the counter **232** (for example, the clock terminal). The selector **231** connects the output terminal C to a selected one of the input terminals A and B in response to the mode selection signal SMS, to select the signal to be input into the counter **232**. Specifically, if the mode selection signal SMS indicates the PFM mode, the selector **231** connects the input terminal A to the output terminal C, to feed the comparison result signal Vout**11** from the comparator **220** to the counter **232**. If the mode selection signal SMS indicates the dual sampling read-out mode, the selector **231** connects the input terminal B to the output terminal C, to feed the clock pulse CP having a predetermined frequency to the counter **232**. The counter **232** counts the first comparison result signal Vout**11** from the comparator **220** in the PFM mode and latches the second comparison result signal Vout**12** in synchronization with the clock pulse CP in the dual sampling read-out mode.

(35) In the first embodiment, the PFM mode is selected under a high illuminance condition and in an exposure period PEXP following resetting of the floating diffusion FD**11** and the photodiode PD**11** serving as the photoelectric conversion element. The PFM mode is followed by the dual sampling read-out mode. In the PFM mode, in the exposure period PEXP, overflowing of the charges from the photodiode PD**11** to the floating diffusion FD**11**, self-resetting of the floating diffusion FD**11** resulting from the feedback resetting through the comparison result signal Vout from the comparator **220**, and counting of the number of times the floating diffusion FD**11** is reset (the reset frequency of the floating diffusion FD**11**) by the selector counter circuit **230** are performed repeatedly. In the present embodiment, the number of times the floating diffusion FD**11** is reset is counted to obtain the amount of actually generated charges. If pulse light or other sorts of abnormal light is incident on a solid-state imaging device during an exposure period, the solid-state imaging device may produce erroneously detected signals. The present embodiment can avoid such errors.

(36) In the first embodiment, the reading part **70** can perform first conversion gain mode reading and second conversion gain mode reading in a read-out period. The reading part **70** reads pixel signals with a first conversion gain corresponding to a first capacitance in the first conversion gain mode reading, and with a second conversion gain corresponding to a second capacitance (different from the first capacitance) in the second conversion gain mode reading. In other words, the solid-state imaging device **10** of the first embodiment switches, in a read-out period for outputting signals, the mode between the first conversion gain (for example, high conversion gain: HCG) mode and the second conversion gain (low conversion gain: LCG) mode in the respective pixels for the charges (electrons) produced by photoelectric conversion in a single exposure period (integration period), so that the solid-state imaging device **10** can output both of bright and dark signals and thus achieve an increased dynamic range.
(37) The following first describes in detail the specific example configuration of the readable pixel **210** of the solid-state imaging device **10** and then an example configuration of a pinned diode (PPD) part and an overflow path. In the following description, the readable pixel **210** has an LOFIC configuration, for example.
(38) <Specific Example Configuration of Readable Pixel **210**>
(39) FIG. **5** is a circuit diagram showing an example configuration of the readable pixel of the pixel circuit relating to the first embodiment of the present invention.
(40) In the pixel part **20**, the pixel circuits **200** are arranged in a two-dimensional matrix comprised of N rows and M columns, and each pixel circuit **200** includes the readable pixel **210** constituted by a photodiode (photoelectric conversion element) and an in-pixel amplifier.
(41) As shown in, for example, FIG. **5**, the readable pixel **210** includes a photodiode PD**11** serving as a photoelectric conversion element, a transfer transistor TG**11**-Tr serving as a charge transfer gate part (a transfer element), a reset transistor RST**11**-Tr serving as a reset element, a self-reset reset transistor RST**12**-Tr, an AND gate AD**11**, a storage transistor SG**11**-Tr serving as a storage element, and a storage capacitor CS**11** serving as a storage capacitance element. In the first embodiment, one of the transistors constituting the differential transistor pair of the comparator **220** also serves as a source follower transistor SF**11**-Tr serving as a source follower element, which will be described below.
(42) The photodiode PD**11** generates signal charges (electrons) in an amount determined by the amount of the incident light and stores the same. A description will be hereinafter given of a case where the signal charges are electrons and each transistor is an n-type transistor. However, it is also possible that the signal charges are holes or each transistor is a p-type transistor.
(43) In each readable pixel **210**, the photodiode (PD) is a pinned photodiode (PPD). The substrate surface for forming the photodiode (PD) has a surface level due to dangling bonds or other defects. Therefore, a lot of charges (dark current) are generated due to heat energy, as a result of which the signals fail to be read out correctly. The pinned photodiode (PPD) has the charge storage part buried in the substrate, thereby reducing mixing of the dark current into signals.
(44) The transfer transistor TG**11**-Tr is connected between the pinned photodiode (PPD) and the floating diffusion FD**11** and controlled through a control signal TG. The transfer transistor TG**11**-Tr remains selected and in the conduction state during a period in which the control signal TG is at the high (H) level, to transfer to the floating diffusion FD**11** the charges (electrons) produced by photoelectric conversion by the photodiode PD**11** and then stored in the storage node ND**10**.
(45) In the example shown in FIG. **5**, the reset transistor RST**11**-Tr is connected between a power supply potential VAAPIX and the storage transistor SG**11**-Tr connected to the floating diffusion FD**11**, and controlled through a control signal RST. The reset transistor RST**11**-Tr remains selected and in the conduction state in the period in which the control signal RST is at the H level, to reset the floating diffusion FD**11** to the power supply potential VAAPIX while the storage transistor SG**11**-Tr is in the conduction state.
(46) In the example shown in FIG. **5**, the self-reset reset transistor RST**12**-Tr is connected between

the power supply potential VAAPIX and the storage transistor SG**11**-Tr connected to the floating diffusion FD**11** and in parallel with the reset transistor RST**11**-Tr. The self-reset reset transistor RST**12**-Tr is controlled through an output control signal SRST from the AND gate AD**11**. The self-reset reset transistor RST**12**-Tr remains selected and in the conduction state in the period in which the control signal SRST is at the H level, to reset the floating diffusion FD**11** to the power supply potential VAAPIX while the storage transistor SG**11**-Tr is in the conduction state.
(47) In the first embodiment, the self-resetting is performed in the PFM mode in the following manner. The self-reset reset transistor RST**12**-Tr and storage transistor SG**11**-Tr remain in the conduction state, so that the floating diffusion FD**11** and the storage capacitor CS**11** are reset.
(48) In the first embodiment, the floating diffusion FD**11** is selectively connected to the storage capacitor CS**11** through the storage transistor SG**11**-Tr, so that the capacitance of the floating diffusion FD**11** is changed between a first capacitance and a second capacitance, so that the conversion gain can be switched between a first conversion gain (high conversion gain: HCG) corresponding to the first capacitance and a second conversion gain (low conversion gain: LCG) corresponding to the second capacitance.
(49) For example, the source of the storage transistor SG**11**-Tr is connected to the floating diffusion FD**11**. The storage capacitor CS**11** has a first electrode EL**1** connected to the reference potential VSS (e.g., the ground potential GND) and a second electrode EL**2** connected to the drain of the storage transistor SG**11**-Tr serving as a capacitance connection node ND**11**. The storage transistor SG**11**-Tr is controlled by a control signal SG applied to the gate thereof through a control line. The storage transistor SG**11**-Tr remains selected and in the conduction state in the period in which the control signal SG is at the H level, to connect the floating diffusion FD**11** and the storage capacitor CS**11**.
(50) The first conversion gain (high conversion gain: HCG) signal read-out operation in the dual sampling read-out mode is performed with the storage transistor SG**11**-Tr remaining in the non-conduction state so that the charges of the floating diffusion FD**11** are separated from the charges of the storage capacitor CS**11**. The second conversion gain (the low conversion gain: LCG) signal read-out operation in the dual sampling read-out mode is performed with the storage transistor SG**11**-Tr remaining in the conduction state so that the charges of the floating diffusion FD**11** and the charges of the storage capacitor CS**11** are combined.
(51) In the example shown in FIG. **5**, the storage transistor SG**11**-Tr is connected between the floating diffusion FD**11** and the reset transistors RST**11**-Tr and RST**12**-Tr, and the storage capacitor CS**11** is connected between the connection node ND**11** of the storage transistor SG**11**-Tr and the reference potential VSS. The manner of connecting these components is not limited to such. For example, the reset transistor RST**11**-Tr and storage transistor SG**11**-Tr may be individually and directly connected to the floating diffusion FD**11**.
(52) The AND gate AD**11** has input terminals one of which is connected to the feeding line of a reset count signal RCNT and the other of which is connected to the feeding line of the comparison result signal Vout from the comparator **220**, and the output terminal of the control signal SRST is connected to the gate of the self-reset reset transistor RST**12**-Tr. The reset count signal RCNT is in synchronization with the mode selection signal SMS. If the mode selection signal SMS is at the high level to indicate the PFM mode period, the reset count signal RCNT is fed at the high level for the same period of time. If the mode selection signal SMS is at the low level to indicate the dual sampling read-out mode, the reset count signal RCNT is fed at the low level for the same period of time. The AND gate AD**11** calculates a logical AND between the reset count signal RCNT, which is fed at the high level in the PFM mode period, and the comparison result signal Vout from the comparator **220**. The control signal SRST is thus output at the high level during the period in which the comparison result signal Vout is at the high level. As a result, while the control signal SRST is at the high level, the self-reset reset transistor RST**12**-Tr remains in the conduction state.
(53) In the first embodiment, the source follower transistor SF**11**-Tr serving as the source follower

element also serves as one of the transistors of the differential transistor pair of the comparator **220**. The following now describes a specific example configuration of the comparator **220**.
(54) <Example Configuration of Comparator **220**>
(55) The comparator **220** relating to the first embodiment has a differential transistor pair and an active load circuit. The differential transistor pair is constituted by transistors a first one of which receives at the gate thereof signal voltage VSL and a second one of which receives at the gate thereof fixed voltage serving as the first reference signal Vref**11** or a ramp signal RAMP serving as the second reference signal Vref**12**. The differential transistor pair is configured to compare the signal voltage VSL against the reference voltage Vref or ramp signal RAMP The active load circuit is configured to form a current mirror when connected to the drain side of the first transistor and the drain side of the second transistor.
(56) FIG. **6** is a circuit diagram showing an example configuration of the comparator relating to the first embodiment.
(57) The comparator **220** in FIG. **6** includes PMOS transistors PT**21**, PT**22**, NMOS transistors NT**21** and NT**22**, a current source **121**, and nodes ND**21** and ND**22**.
(58) The source of the PMOS transistor PT**21** and the source of the PMOS transistor PT**22** are connected to each other, and the connection node therebetween is connected to the power supply potential VDD. The source of the NMOS transistor NT**21** and the source of the NMOS transistor NT**22** are connected to each other, and the connection node therebetween is connected to the current source **121** connected to the reference potential VSS.
(59) The drain of the PMOS transistor PT**21** is connected to the drain of the NMOS transistor NT**21**, and the connecting point therebetween forms a node ND**21**. The drain of the PMOS transistor PT**22** is connected to the drain of the NMOS transistor NT**22**, and the connecting point therebetween serves as a node ND**22**. The node ND**21** is connected to the gates of the PMOS transistors PT**21** and PT**22**, and the node ND**22** is connected to the output node ND**221**.
(60) Among these constituents, the NMOS transistors NT**21** and NT**22** the sources of which are connected to each other constitute a differential transistor pair **221**. The NMOS transistor NT**21** constitutes a first one of the transistors (**221-1**), and the NMOS transistor NT**22** constitutes a second one of the transistors (**221-2**). The voltage signal VSL is fed to the gate of the NMOS transistor NT**21** constituting the first transistor **221-1**, and the fixed voltage serving as the first reference signal Vref**11** or the ramp signal RAMP serving as the second reference signal Vref**12** is fed to the gate of the NMOS transistor NT**22** constituting the second transistor **221-2**.
(61) The active load circuit **222**, which is configured to form a current mirror with the PMOS transistors PT**21** and PT**22**, is arranged on the drain side of the NMOS transistor NT**21** constituting the first transistor and on the drain side of the NMOS transistor NT**22** constituting the second transistor.
(62) In the comparator **220** described above, the connection node ND**22** between the drain of the NMOS transistor NT**22** serving as the second transistor and the drain of the PMOS transistor PT**22** constituting the active load circuit **222** is connected to the output node ND**221** for outputting the comparison result signal Vout. Stated differently, the comparator **220** outputs the comparison result signal Vout from the drain side of the NMOS transistor NT**22** serving as the second transistor.
(63) For example, the comparator **220** compares the voltage signal VSL against the first reference signal Vref**11** or second reference signal Vref**12**. If the voltage signal VSL becomes equal to the first or second reference signal Vref **11** or Vref**12**, the output level of the comparison result signal Vout is inverted from the non-active level (for example, low level) to the active level (high level).
(64) The source follower transistor SF**11**-Tr, which also serves as one of the differential transistors of the comparator **220**, outputs a read-out signal (voltage signal) of a column output generated by converting the charges at the floating diffusion FD**11** into a voltage signal with a gain determined by the capacitance. These operations are performed concurrently and in parallel in the pixels in a given one of the rows, because the gates of the transfer transistors TG**11**-Tr, reset transistors

RST**11**-Tr and RST**12**-Tr, and storage transistors SG**1**-Tr in each of the rows are connected together.
(65) Since the pixel part **20** includes the pixel circuits **200** arranged in N rows and M columns, N control lines are provided for each of the control signals, and M vertical signal lines are provided. In FIG. **3**, the control lines for each row are represented as one row-scanning control line.
(66) The vertical scanning circuit **30** drives the pixels in shutter and read-out rows through the row-scanning control lines under control of the timing control circuit **60**. Further, the vertical scanning circuit **30** outputs, according to address signals, row selection signals for row addresses of the reading rows from which signals are read out and the shutter rows in which the charges stored in the photodiodes PD**11** are reset.
(67) The reading circuit **40** includes a plurality of column signal processing circuits (not shown) arranged corresponding to the column outputs of the pixel part **20**, and the reading circuit **40** may be configured such that the plurality of column signal processing circuits can perform column parallel processing.
(68) The horizontal scanning circuit **50** scans the signals processed in the plurality of column signal processing circuits of the reading circuit **40**, transfers the signals in a horizontal direction, and outputs the signals to a signal processing circuit (not shown).
(69) The timing control circuit **60** generates timing signals required for signal processing in the pixel part **20**, the vertical scanning circuit **30**, the reading circuit **40**, the horizontal scanning circuit **50**, and the like.
(70) The above description has outlined the configurations and functions of the parts of the solid-state imaging device **10**. The following now describes in detail the configuration and function of the readable pixel according to the first embodiment.
(71) <Specific Example Configuration of Readable Pixel **210**>
(72) FIG. **7** is a simplified sectional view showing an example configuration of the readable pixel according to the first embodiment of the present invention. The readable pixel described herein includes a pinned photodiode (PPD) and is denoted by the reference sign **2000**.
(73) The readable pixel **2000** shown in FIG. **7** includes a semiconductor substrate (hereinafter referred to simply as "the substrate") **2100** having a first substrate surface **2110** (e.g., back surface) to be irradiated with light L and a second substrate surface **2120** (front surface) facing away from the first substrate surface **2110**. The readable pixel **2000** includes: a photoelectric conversion part **2200** serving as the photodiode PD**11** which includes a first conductivity type (n-type in this embodiment) semiconductor layer (n layer) **2210** buried in the substrate **2100** and is configured to perform photoelectric conversion of received light and store charges; and a second conductivity type (p-type in this embodiment) semiconductor layer **2300** formed at least on a side portion of the n layer (the first conductivity type semiconductor layer) **2210** of the photoelectric conversion part **2200**. The n-layer (first conductivity type semiconductor layer) **2210** of the photoelectric conversion part **2200** exhibits a gradient in concentration such that the impurity concentration of the n-type ions gradually increases from the first substrate surface **2110** side to the second substrate surface **2120** side.
(74) Further, the readable pixel **2000** includes: the transfer transistor TG**11**-Tr that transfers the charges stored in the photoelectric conversion part **2200**; the floating diffusion FD**11** to which the charges are transferred through the transfer transistor TG**11**-Tr; the storage transistor SG**11**-Tr with the source connected to the floating diffusion FD**11**; and the storage capacitor CS**11** serving as the storage capacitance element that stores the charges received from the floating diffusion FD**11** via the drain side of the storage transistor SG**11**-Tr. In the readable pixel **2000**, the storage capacitor CS**11** serving as a storage capacitance element is formed on the second substrate surface **2120** side so as to spatially overlap with the photoelectric conversion part **2200** in the direction perpendicular to the substrate surface (the direction Z in the orthogonal coordinate system shown).
(75) Further, the photoelectric conversion part **2200** includes a second conductivity type

semiconductor region (a p+ region) **2230** formed on the surface of the n layer (the first conductivity type semiconductor layer) **2220** on the second substrate surface **2120** side. The second conductivity type semiconductor region (the p+ region) **2230** contains a higher concentration of impurities than the p layer (the second conductivity type semiconductor layer) **2300** on the side portion of the n layer. The storage capacitor CS**11** serving as the storage capacitance element uses, as the first electrode EL**1** thereof, the n+ region (the first conductivity type semiconductor region) **2240** formed in the surface of the p layer (the second conductivity type semiconductor layer) **2300** on the second substrate surface **2120** side.
(76) In the first embodiment, the storage capacitor CS**11** serving as the storage capacitance element includes the first electrode EL**1** and a second electrode EL**2**. The first electrode EL**1** is formed of the n+ region (the first conductivity type semiconductor region) **2240** formed in the surface of the second substrate surface **2120** of the substrate **2100**, and the second electrode EL**2** is formed above the second substrate surface **2120** so as to be opposed at a distance to the first electrode EL**1** in the direction perpendicular to the substrate surface.
(77) In the first embodiment, a flat layer **2250** is formed on the surfaces of the n layer (the first conductivity type semiconductor layer) **2210** of the photoelectric conversion part **2200** and the p layer (the second conductivity type semiconductor layer) **2300** on the first substrate surface **2110** side. A color filter part CF is formed on the light incidence side of the flat layer **2250**, and further, a microlens MCL is formed on the light incidence side of the color filter part so as to correspond to the photoelectric conversion part **2200** serving as the photodiode PD**11** and the p layer (the second conductivity type semiconductor layer) **2300**.
(78) In the first embodiment, the p layer (the first conductivity type semiconductor layer) **2320** in the right region in the drawing of the second substrate surface **2120** of the substrate **2100** contains the transfer transistor TG**11**-Tr, the floating diffusion FD**11**, and the storage transistor SG**11**-Tr.
(79) The floating diffusion FD**11** is formed in the surface of the second substrate surface **2120** of the substrate **2100**, in the form of an n+ region (the first conductivity type semiconductor region) **2330** containing a higher concentration of impurities than the n layers (the first conductivity type semiconductor layers) **2210** and **2220** of the photoelectric conversion part **2200**.
(80) The capacitance connection node ND**11** of the storage transistor SG**11**-Tr for capacitance connection with the storage capacitor CS**11** is formed in the surface of the second substrate surface **2120** of the substrate **2100**, in the form of an n+ region (the first conductivity type semiconductor region) **2340** containing a higher concentration of impurities than the n layers (the first conductivity type semiconductor layers) **2210** and **2220** of the photoelectric conversion part **2200**. The n+ region **2340** serving as the node ND**11** is connected to the second electrode EL**2** of the storage capacitor CS**11** via a wiring layer WR**1**.
(81) The transfer transistor TG**11**-Tr includes a gate electrode **2510** disposed above the second substrate surface **2120** of the substrate **2100** between the p+ region (the second conductivity type semiconductor region) **2230** and the n+ region (the first conductivity type semiconductor region) **2330** serving as the floating diffusion FD**11**.
(82) The storage transistor SG**11**-Tr includes a gate electrode **2520** disposed above the second substrate surface **2120** of the substrate **2100** between the n+ region (the first conductivity type semiconductor region) **2330** serving as the floating diffusion FD**11** and the n+ region (the first conductivity type semiconductor region) **2340** serving as the capacitance connection node ND**11**.
(83) In the first embodiment, a buried overflow path **2600** is formed and connected to the upper side (the surface facing the p+ layer **2230**) of the n layer **2220** of the photoelectric conversion part **2200**. The buried overflow path **2600** is configured to transfer the overflow charges from the photoelectric conversion par t **2200** to the floating diffusion FD**11**, further to the capacitive connection node ND**11**, which is connected to the storage capacitor CS**11**. The buried overflow path **2600** is formed such that (i) the lower layer of the channel forming region under the gate electrode **2510** of the transfer transistor TG**11**-Tr, (ii) the lower layer of the n+ region (the first

conductivity type semiconductor region) **2330** serving as the floating diffusion FD**11**, (iii) the lower layer of the channel forming region under the gate electrode **2520** of the storage transistor SG**11**-Tr, and (iv) the lower layer of the n+ region (the first conductivity type semiconductor region) **2340** serving as the capacitance connection node ND**11** are in communication with each other. The buried overflow path **2600** is formed by an n− layer with a low impurity concentration in the n layer **2220** of the photoelectric conversion part **2200**.

(84) As described above, in the readable pixel **2000** relating to the first embodiment, the buried overflow path **2600** is formed and connected to the upper side (the surface facing the p+ layer **2230**) of the n layer **2220** of the photoelectric conversion part **2200**. The buried overflow path **2600** is configured to transfer the overflow charges from the photoelectric conversion part **2200** to the floating diffusion FD**11**, further to the capacitance connection node ND**11**, which is connected to the storage capacitor CS**11**. Therefore, through the entire exposure period, the transfer transistor TG**11**-Tr remains off (in the non-conduction state), so that dark current is prevented at the silicon interface under the transfer transistor TG**11**-Tr while the charges can still overflow.

(85) In the readable pixel **2000** of the first embodiment, the storage capacitor CS**11** serving as a storage capacitance element faces the second substrate surface **2120** and includes the first electrode EL**1** and the second electrode EL**2**. The first electrode EL**1** is formed of the n+ region (the first conductivity type semiconductor region) **2240** formed in the surface of the second substrate surface **2120** of the substrate **2100**, and the second electrode EL**2** is formed above the second substrate surface **2120** so as to be opposed at a distance to the first electrode EL**1** in the direction perpendicular to the substrate surface. The first electrode EL**1** and the second electrode EL**2** are arranged so as to spatially overlap with the photoelectric conversion part **2200** in the direction perpendicular to the substrate surface (the direction Z in the orthogonal coordinate system shown). An increase in the capacitance of the storage capacitor CS**11** may result in a reduced opening of the photodiode PD and thus in a lower sensitivity. The present invention, however, can avoid such drawbacks. The present invention provide for additional advantages. An increase in the light receiving area of the photodiode PD**11** may result in reduction of the area occupied by the storage capacitor CS**11** and eventually in a reduced dynamic range. The present invention can also prevent this issue. As described above, the first embodiment can implement both a high dynamic range and a high sensitivity without a trade-off.

(86) As described above, in the first embodiment, the PFM mode is selected under a high illuminance condition and in the exposure period PEXP, which starts after the floating diffusion FD**11** and the photodiode PD**11** serving as the photoelectric conversion element are reset. The PFM mode is followed by the dual sampling read-out mode.

(87) If the dual sampling read-out mode is selected following the PFM mode, the reading part **70** controls the storage transistor SG**11**-Tr to remain in the non-conduction state, to disconnect the storage capacitor CS**11** from the floating diffusion FD**11**. As a result, the charges in the floating diffusion FD**11** are separated from the charges in the storage capacitor CS**11**, so that the conversion gain is switched to the first conversion gain HCG corresponding to the first capacitance. In a first reset read-out period following the resetting, the reading part **70** performs a first conversion gain reset read-out operation HCGRRD. Specifically, the reading part **70** reads a first read-out reset signal HCGRST (ADC), which is produced through conversion with the first conversion gain corresponding to the first capacitance of the floating diffusion FD**11**, from the source follower transistor SF**11**-Tr serving as the output buffer part and processes the first read-out reset signal HCGRST (ADC) in a predetermined manner. The first reset read-out period is followed by a first transfer period and then a first read-out period, in which the reading part **70** further performs a first conversion gain read-out operation HCGSRD. Specifically, the reading part **70** reads a first read-out signal HCGSIG (ADC), which is produced through conversion with the first conversion gain corresponding to the first capacitance of the floating diffusion FD**11** from the source follower transistor SF**11**-Tr serving as the output buffer part and processes the first read-out signal HCGSIG

(ADC) in a predeterminer manner.
(88) The reading part **70** either holds the reset level and signal level, or performs a CDS operation based on the difference between the reset level and the signal level. After performing the first conversion gain read-out operation HCGSRD, the reading part **70** then switches the storage transistor SG**11**-Tr into the conduction state, to connect the storage capacitor CS**11** to the floating diffusion FD**11**. As a result, the charges in the floating diffusion FD**11** are combined with the charged in the storage capacitor CS**11**, so that the conversion gain is switched to the second conversion gain LCG corresponding to the second capacitance. The first read-out period is followed by a second transfer period and then a second read-out period, in which the reading part **70** further performs a second conversion gain read-out operation LCGSRD. Specifically, the reading part **70** reads a second read-out signal LCGSIG (ADC), which is produced through conversion with the second conversion gain corresponding to the second capacitance of the floating diffusion FD**11** from the source follower transistor SF**11**-Tr serving as the output buffer part and processes the second read-out signal LCGSIG (ADC) in a predetermined manner. After the reset transistor RST**11**-Tr resets the floating diffusion FD**11**, the reading part **70** performs a second conversion gain reset read-out operation LCGRRD. Specifically, the reading part **70** reads a second read-out reset signal LCGRST (ADC), which is produced through conversion with the second conversion gain LCG corresponding to the second capacitance of the floating diffusion FD**11**, from the source follower transistor SF**11**-Tr serving as the output buffer part and processes the second read-out reset signal LCGRST (ADC) in a predetermined manner.
(89) The reading part **70** either holds the reset level and signal level, or performs a DDS operation based on the difference between the reset level and the signal level. In the first embodiment, the reset level of the source follower is obtained after the signal level of the source follower is obtained and by resetting the floating diffusion FD**11**. Therefore, if a subject has high illuminance, the reset level is an imperfect difference signal (double data sampling (DDS) signal or delta reset sampling (DRS) signal) that has no correlation with reset noise. Such reset noise may stand out in normal CMOS image sensors. According to the read-out scheme of the first embodiment, however, the DDS signal is not used under a low illuminance condition since it is produced under a high illuminance condition and read through the source follower reading. The reset noise thus does not stand out as it is concealed by signal shot noise and thus becomes less invisible.
(90) The reading part **70** can perform the first conversion gain reset read-out operation HCGRRD in the exposure period PEXP. Under a very low illuminance condition, the reading part **70** performs the second conversion gain reset read-out operation LCGRRD and second conversion gain read-out operation LCGSRD, as the normal read-out operation. Under a medium illuminance condition, the reading part **70** performs the first conversion gain reset read-out operation HCGRRD, first conversion gain read-out operation HCGSRD, second conversion gain read-out operation LCGSRD, and second conversion gain reset read-out operation LCGRRD.
(91) As described above, the first embodiment includes the storage transistor SG**11**-Tr and storage capacitor CS**11**, so that the capacitance of the floating diffusion FD**11** can be switched between the first capacitance and the second capacitance, to change the conversion gain between the first conversion gain corresponding to the first capacitance (for example, high conversion gain: HCG) and the second conversion gain corresponding to the second capacitance (for example, low conversion gain: LCG). With such configurations, the full well capacity (FWC) is small when the conversion gain is set at the high conversion gain (HCG) and large when the conversion gain is set at the low conversion gain (LCG).
(92) The following now describes, as an example, a sequence of operations performed for reading on the pixel circuit in the solid-state imaging device relating to the first embodiment.
(93) FIG. **8** is a timing chart including waveforms (A) to (H) to illustrate an example sequence of operations for reading performed in the PFM mode and dual sampling read-out mode on the pixel circuit under a high luminance condition in the solid-state imaging device relating to the first

embodiment of the present invention. FIG. **9** includes views (A) to (N) to illustrate a sequence of operations and potential transition to explain the operations performed in the PFM mode and dual sampling read-out mode under a high illuminance condition on the pixel circuit of the solid-state imaging device relating to the first embodiment of the present invention. FIG. **10** is a timing chart including waveforms (A) to (H) to illustrate an example sequence of operations for reading performed in a PFM mode and a dual sampling read-out mode on the pixel circuit under a low illuminance condition in the solid-state imaging device relating to the first embodiment of the present invention. FIG. **11** includes views (A) to (N) to illustrate a sequence of operations and potential transition to explain the operations performed in a PFM mode and a dual sampling read-out mode under a low illuminance condition on the pixel circuit of the solid-state imaging device relating to the first embodiment of the present invention.

(94) In FIGS. **8** and **10**, the waveform (A) shows the reset count signal RCNT in synchronization with the mode selection signal SMS and indicating the PFM mode period or dual sampling read-out mode period, the waveform (B) shows the control signal RST for the reset transistor RST**11**-Tr, the waveform (C) shows the control signal SG for the storage transistor SG**11**-Tr, the waveform (D) shows the control signal TG for the transfer transistor TG**11**-Tr, the waveform (E) shows the potential of the floating diffusion FD**11**, the waveform (F) shows the output voltage SFout from the source follower transistor SF**11**-Tr and the reference signal Vref (**1**, **2**) for the comparator **220**, the waveform (G) shows the digital output voltage Vout from the comparator **220**, and the waveform (H) shows the clock signal CLK used in the dual sampling read-out mode.

(95) <Operations Performed Under High Illuminance Condition>

(96) Under a high illuminance condition, prior to the PFM mode, the control signal RST is set at the high level to control the reset transistor RST**11**-Tr to remain in the conduction state. After this, the control signal TG is set at the high level to control the transfer transistor TG**11**-Tr to remain in the conduction state, so that the floating diffusion FD**11** is connected to the photodiode PD**11**. The control signal SG is then switched from the low level to the high level, to switch the storage transistor SG**11**-Tr from the non-conduction state to the conduction state. This resets the photodiode PD**11** and floating diffusion FD**11** to the fixed potential VAAPIX. Stated differently, global resetting is performed (see the view (A) in FIG. **9**). The control signal TG is then switched from the high level to the low level to switch the transfer transistor TG**11**-Tr from the conduction state to the non-conduction state.

(97) At the time when the transfer transistor TG**11**-Tr is switched from the conduction state to the non-conduction state, the exposure period PEXP starts (the view (B) in FIG. **9**). After the exposure period PEXP starts, the control signal RST is switched from the high level to the low level to switch the reset transistor RST**11**-Tr from the conduction state into the non-conduction state, the reset count signal RCNT in synchronization with the mode selection signal SMS activates the PFM mode.

(98) In the PFM mode under a high illuminance condition, if the voltage signal corresponding to the charges held in the floating diffusion FD**11** reaches the reference signal Vref**11**, the comparison result signal Vout from the comparator **220** is applied to perform feedback resetting FRST**11** to control the reset transistor RST**12**-Tr to remain in the conduction state for a predetermined period of time, so that the floating diffusion FD**11** is self-reset (see the views (C), (D) and (E) in FIG. **9**). The number of times the floating diffusion FD**11** is reset (the reset frequency of the floating diffusion FD**11**) is counted by the selector counter circuit **230** in response to the comparison result signal Vout from the comparator **220**. In the PFM mode under a high illuminance condition, in the exposure period PEXP, overflowing of the charges from the photodiode PD**11** to the floating diffusion FD**11**, self-resetting of the floating diffusion FD**11** resulting from the feedback resetting through the comparison result signal Vout from the comparator **220**, and counting of the number of times the floating diffusion FD**11** is reset (the reset frequency of the floating diffusion FD**11**) by the selector counter circuit **230** are performed repeatedly (see the views (C), (D) and (E) in FIG. **9**).

(99) To end the PFM mode and start the dual sampling (LOFIC) read-out mode, the reset count signal RCNT in synchronization with the mode selection signal SMS is switched from the high level to the low level. In parallel, the control signal SG is switched from the high level to the low level to place the storage transistor SG**11**-Tr into the non-conduction state, so that the storage capacitor CS**11** is disconnected from the floating diffusion FD**11**.
(100) As a result, the charges in the floating diffusion FD**11** and the charges in the storage capacitor CS**11** are separated, so that the gain of the floating diffusion FD**11** is switched to the first conversion gain HCG corresponding to the first capacitance.
(101) In the first reset read-out period following the resetting, the first conversion gain reset read-out operation HCGRRD is performed. Specifically, the first read-out reset signal HCGRST (ADC), which is produced through conversion with the first conversion gain HCG corresponding to the first capacitance of the floating diffusion FD**11**, is read from the source follower transistor SF**11**-Tr (see the view (G) in FIG. **9**) and processed in a predetermined manner. The first reset read-out period is followed by the first transfer period, in which the control signal TG is switched to the high level to keep the transfer transistor TG**11**-Tr in the conduction state, so that the charges stored in the photodiode PD**11** are transferred to the floating diffusion FD**11**. After the first transfer period, the control signal TG is switched to the low level to switch the transfer transistor TG**11**-Tr into the non-conduction state.
(102) The first transfer period is followed by the first read-out period, in which the first conversion gain read-out operation HCGSRD is performed. Specifically, the first read-out signal HCGSIG (ADC), which is produced through conversion with the first conversion gain corresponding to the first capacitance of the floating diffusion FD**11**, is read from the source follower transistor SF**11**-Tr and processed in a predetermined manner.
(103) The reset level HCGRSTADC and the signal level HCGSIGADC are held, or a CDS operation is performed based on the difference between the reset level and the signal level.
(104) After the first conversion gain read-out operation HCGSRD, the control signal SG is switched from the low level to the high level to place the storage transistor SG**11**-Tr into the conduction state, so that the storage capacitor CS**11** is connected to the floating diffusion FD**11**. As a result, the charges in the floating diffusion FD**11** and the charges in the storage capacitor CS**11** are combined, so that the gain of the floating diffusion FD**11** is switched to the second conversion gain LCG corresponding to the second capacitance.
(105) The second read-out period is followed by the second transfer period, in which the control signal TG is switched to the high level to keep the transfer transistor TG**11**-Tr in the conduction state, so that the charges stored in the photodiode PD**11** are transferred to the floating diffusion FD**11**. After the second transfer period, the control signal TG is switched to the low level to switch the transfer transistor TG**11**-Tr into the non-conduction state. The second read-out period is followed by the second transfer period and then the second read-out period, in which the second conversion gain read-out operation LCGSRD is performed. Specifically, the second read-out signal LCGSIG (ADC), which is produced through conversion with the second conversion gain LCG corresponding to the second capacitance of the floating diffusion FD**11**, is read from the source follower transistor SF**11**-Tr (see the view (L) in FIG. **9**) and processed in a predetermined manner.
(106) The second transfer period is followed by the second reset read-out period, in which the second conversion gain reset read-out operation LCGRRD is performed. Specifically, the second read-out reset signal LCGRST (ADC), which is produced through conversion with the second conversion gain LCG corresponding to the second capacitance of the floating diffusion FD**11**, is read from the source follower transistor SF**11**-Tr and processed in a predetermined manner.
(107) The reset level LCGRSTADC and the signal level LCGSIGADC are held, or a DDS operation is performed based on the difference between the reset level LCGRSTADC and the signal level LCGSIGADC.
(108) <Operations Performed under Low Illuminance Condition>

(109) Under a low illuminance condition, the reference signal Vref**11** corresponding to the charges held in the floating diffusion FD**11** is never reached in the PFM mode the voltage signal corresponding to the charges held in the floating diffusion FD**11** never reaches the reference signal Vref**11** in the PFM mode. Therefore, in the PFM mode under a low illuminance condition, in the exposure period PEXP, overflowing of the charges from the photodiode PD**11** to the floating diffusion FD**11**, self-resetting of the floating diffusion FD**11** resulting from the feedback resetting through the comparison result signal Vout from the comparator **220**, and counting of the number of times the floating diffusion FD**11** is reset (the reset frequency of the floating diffusion FD**11**) by the selector counter circuit **230** are never performed repeatedly.
(110) The series of operations performed in the dual sampling (LOFIC) read-out mode under a low illuminance condition are the same as those performed under a high illuminance condition, which have been described in the above. They are thus not described in detail here.
(111) As described above, the pixel circuit **200** relating to the first embodiment includes: the readable pixel **210** for performing photoelectric conversion at the photodiode PD**11** serving as the photoelectric conversion element and producing a readable signal corresponding to an illuminance condition of incident light; the comparator **220** for comparing the voltage signal SFout read from the readable pixel **210** against the reference signal Vref and outputting the comparison result signal Vout corresponding to the result of the comparison; the selector counter circuit **230** for selecting the count clock for the counter in response to the comparison result signal Vout from the comparator **220** and performing counting in synchronization with the selected clock; and the memory circuit **240** for storing digital data produced by the selector counter circuit **230** based on the comparison result signal Vout from the comparator **220**. The pixel circuit **200** includes the overflow path **2600** for allowing the charges overflowing from the photodiode PD**11** to overflow to the floating diffusion FD**11**, further to the connection node connected to the storage capacitor CS**11**. The readable pixel **210**, comparator **220** and selector counter circuit **230** operate in the pulse frequency modulation (PFM) mode under a high illuminance condition and in the dual sampling read-out (for example, LOFIC) mode under a low illuminance condition using the charges stored in the photoelectric conversion element and the overflow charges. Specifically, under a high illuminance condition or in the PFM mode, if the voltage signal corresponding to the charges held in the floating diffusion FD**11** reaches the full well voltage of the photodiode PD**11** serving as the photoelectric conversion element, the comparison result signal Vout from the comparator **220** is applied to perform the feedback resetting FRST**11** to allow the floating diffusion FD**11** to perform self-resetting. The selector counter circuit **230** counts the number of times the floating diffusion FD**11** is reset (the reset frequency of the floating diffusion FD**11**) in response to the comparison result signal Vout from the comparator **220**. Under a low illuminance condition, the dual sampling read-out mode (LOFIC) mode is selected that uses the dual gain scheme based on the charges stored in the photodiode PD**11** serving as the photoelectric conversion element and the overflow charges.
(112) Since the buried overflow path extends at least from the photodiode PD**11** to the region where the floating diffusion FD**11** is formed in the first embodiment, the transfer transistor TG**11**-Tr remains turned-off (in the non-conduction state) through the entire exposure period to allow the charges to overflow while dark current is prevented at the silicon interface under the transfer transistor TG**11**-Tr. Stated differently, low signals can be read even while the saturation is reached without compromising the sensitivity and full well in the first embodiment. In addition, the first embodiment can achieve reduced pixel size.
(113) The first embodiment can also provide for enhanced dynamic range by performing reading in a predetermined mode, while accomplishing a small pixel size. The first embodiment is capable of substantially achieving an increased dynamic range and a raised frame rate. The first embodiment produces further effects. Very-high-dynamic-range signals can be read with a linear response in the PFM read-out mode as shown in FIG. **12**, and high-sensitivity/low-noise signals can be read in the

HCG read-out mode. The first embodiment can prevent the uneven full well capacities from being translated into uneven signals at the combining point between the pixels. In the first embodiment, the number of times the floating diffusion FD**11** is reset is counted to obtain the amount of actually generated charges. If pulse light or other sorts of abnormal light is incident on a solid-state imaging device during an exposure period, the solid-state imaging device may produce erroneously detected signals. The present embodiment can avoid such errors.

Second Embodiment

(114) FIG. **13** is a circuit diagram showing an example configuration of a readable pixel of a pixel circuit relating to a second embodiment of the present invention. FIG. **14** is a simplified sectional view showing an example configuration of a charge integrating and transferring system including a shutter gate transistor, or the main part of the pixel circuit relating to the second embodiment of the present invention. FIG. **15** is a timing chart including waveforms (A) to (H) to illustrate an example sequence of operations performed for reading in the PF mode and dual sampling read-out mode on the pixel circuit under a high luminance condition in the solid-state imaging device relating to the second embodiment of the present invention. FIG. **16** is a timing chart including waveforms (A) to (I) to illustrate an example sequence of operations performed for reading in the PFM mode and dual sampling read-out mode on the pixel circuit under a low illuminance condition in the solid-state imaging device relating to the second embodiment of the present invention.

(115) The pixel circuit **200**A of the solid-state imaging device **10**A relating to the second embodiment differs from the pixel circuit **200** of the solid-state imaging device **10** relating to the above-described first embodiment in the following points.

(116) The pixel circuit **200**A of the solid-state imaging device **10**A of the second embodiment includes an anti-blooming transistor AB**11**-Tr serving as a gate element connected between the charge storage region in the photodiode PD**11** and a fixed potential (for example, power supply potential) VAAPIX. The anti-blooming transistor AB**11**-Tr is configured to discharge the charges in the photodiode PD**11** to the fixed potential VAAPIX located outside the region of the floating diffusion FD**11**.

(117) In the solid-state imaging device **10**A relating to the second embodiment, the overflow path **2600** is buried as shown in FIG. **14** (and FIG. **7**). Therefore, the overflow barrier of the transfer transistor TG**11**-Tr is lower than the overflow barrier of the anti-blooming transistor AB**11**-Tr. This causes the charges in the photodiode PD**11** to overflow only toward the floating diffusion FD**11**.

(118) <Configuration of Left Separation Layer **2310** in X direction (Column Direction)>

(119) An n+ layer **2350** serving as the drain of the anti-blooming transistor AB**11**-Tr is formed on the p-type separation layer **2310**, which is arranged on the left side in the X direction (column direction) in FIG. **14**, on the second substrate surface **2120** side. On the second substrate surface **2120**, a gate electrode **2540** of the anti-blooming transistor AB**11**-Tr is formed with a gate insulator sandwiched therebetween. Under the transfer transistor TG**11**-Tr, the overflow path **2600** extends from the photodiode PD**11** to the floating diffusion FD**11**, finally to the connection node **2340** connected to the storage capacitor CS**11**. The potential of the overflow path **2600** can be controlled through gate control, for example.

(120) With the above-described structure, if the intensity of the incident light is very high (the amount of the incident light is very large), the charges above the well capacity of the PD overflow into the floating diffusion FD**11** as overflow charges through the overflow path **2600** under the transfer transistor TG**11**-Tr.

(121) In the solid-state imaging device **10**A relating to the second embodiment, the charges stored in the photodiode PD**11** are reset by controlling the anti-blooming transistor AB**11**-Tr to remain in the conduction state for a predetermined period of time, as shown in the waveform (E) in FIGS. **15** and **16**. The exposure period PEXP starts when the control signal AB for the anti-blooming transistor AB**11**-Tr is switched from the high level to the low level.

(122) In other respects, the second embodiment is the same as the first embodiment described

above. The second embodiment can not only produce the same effects as the above-described first embodiment but also produce the following effects.
(123) The second embodiment can prevent mixing of charges (which can result in false signals), which is caused by the signals indicative of the charges exceeding the charges that can be stored on a given photodiode PD (overflow charges) flowing into adjacent pixels.
Third Embodiment
(124) FIG. **17** is a circuit diagram showing an example configuration of a pixel circuit of a solid-state imaging device **10**B relating to a third embodiment of the present invention.
(125) The pixel circuit **200**B of the solid-state imaging device **10**B relating to the third embodiment differs from the pixel circuit **200**A of the solid-state imaging device **10**A relating to the above-described second embodiment in terms of the configuration of a comparator **220**B.
(126) In the pixel circuit **200**B of the solid-state imaging device **10**B relating to the third embodiment, a first input terminal or inversion input terminal (−) of the comparator **220**B receives the voltage signal VSL fed thereto, which is output from the output node of the readable pixel **210**B connected to the source side of the source follower transistor SF**11**-Tr, and a second input terminal or non-inversion input terminal (+) of the comparator **220**B receives the reference signal (referential voltage) Vref (**11**, **12**) fed thereto. The comparator **220**B performs AD conversion (a comparing operation) of comparing the voltage signal VST against the reference signal Vref and outputting a digital comparison result signal SCMP.
(127) The first input terminal or inversion input terminal (−) of the comparator **220**B is connected to a coupling capacitor CC**1**. In this way, the output node of the source follower transistor SF**11**-Tr serving as the output buffer part of the readable pixel **210**B formed on a first substrate is AC coupled to the input part of the comparator **220**B formed on a second substrate, so that the noise can be reduced and high SNR can be achieved when the illuminance is low.
(128) In the comparator **220**B, an auto-zero switch SW-AZ serving as a reset switch is connected between the output terminal and the first input terminal or inversion input terminal (−). This can eliminate the offset at the comparator **220**B.
(129) The comparator **220**B compares the analog signal (the potential VSL) read from the source follower transistor SF**11**-Tr serving as the output buffer part of the readable pixel **210**B against the reference signal Vref**12**, for example, the ramp signal RAMP that linearly changes with a certain gradient or has a slope waveform. During the comparison, a counter (not shown), which is provided for each column as is the comparator **220**B, is operating. The ramp signal RAMP having a ramp waveform and the value of the counter vary in a one-to-one correspondence, so that the voltage signal VSL is converted into a digital signal. Basically, the AD converting part converts a change in voltage, in other words, a change in the reference signal Vref (for example, the ramp signal RAMP) into a change in time, and counts the change in time at certain intervals (with certain clocks). In this way, the AD converting part obtains a digital value. When the analog signal VSL and the ramp signal RAMP (the reference signal Vref) cross each other, the output from the comparator **220**B is inverted, the clock input into the counter is stopped or the suspended clock is input into the counter, and the value (data) of the counter at that timing is saved in the memory part **240**. In this way, the AD conversion is completed.
(130) As described above, the third embodiment can eliminate the offset at the comparator **220**B and also achieve reduced noise and high SNR under a low illuminance condition.
Fourth Embodiment
(131) FIG. **18** is a circuit diagram showing an example configuration of a pixel circuit of a solid-state imaging device **10**C relating to a fourth embodiment of the present invention.
(132) The pixel circuit **200**C of the solid-state imaging device **10**C relating to the fourth embodiment differs from the pixel circuit **200** of the solid-state imaging device **10** relating to the above-described first embodiment in that the specific configuration of a selector counter circuit **230**C is shown.

(133) The pixel circuit **200**C of the solid-state imaging device **10**C relating to the fourth embodiment includes the selector counter circuit **230**C, which is mainly constituted by an up-down (U/D) counter **233**, and a selector **234** for selecting a signal to be input into the clock terminal of the up-down counter **233** based on the comparison result signal output from the comparator **220**.
(134) The selector **234** has a first input terminal A connected to the reference potential VSS, a second input terminal B connected to the feeding line of a clock signal SCLK, and an output terminal C connected to the clock terminal CK of the up-down counter **233**. The clock signal SCLK is set at a fixed potential in the PFM mode and configured as a clock pulse having a predetermined frequency in the dual sampling read-out mode. If the output from the comparator **220** flips (is inverted) in the PFM mode, the selector **234** disconnects the connection between the output terminal C and the first input terminal A and establishes connection between the output terminal C and the second input terminal B. If the output from the comparator **220** flips in the dual sampling read-out mode, the selector **234** disconnects the connection between the output terminal C and the second input terminal B and establishes connection between the output terminal C and the first input terminal A.
(135) In the fourth embodiment, the use of the up-down counter eliminates the necessity of the memory for storing the reset signals HCGRST and LCGRST. The memory circuit **240** is thus only required to include three memory units, i.e., the memory **241** for the PFM, the memory **242** for the HCG read-out signal HCGSIG, and the memory **243** for the LCG read-out signal LCGSIG.
(136) The fourth embodiment thus can not only produce the same effects as the above-described first to third embodiments but also allow the memory to have a reduced size.
Fifth Embodiment
(137) FIG. **19** is a circuit diagram showing an example configuration of a pixel circuit of a solid-state imaging device **10**D relating to a fifth embodiment of the present invention.
(138) The pixel circuit **200**D of the solid-state imaging device **10**D relating to the fifth embodiment differs from the pixel circuit **200** of the solid-state imaging device **10** relating to the above-described first embodiment in terms of the configuration of a selector counter circuit **230**D.
(139) The pixel circuit **200**D of the solid-state imaging device **10**D relating to the fifth embodiment includes the selector counter circuit **230**D, which is mainly constituted by a counter **235** and a latch **236**. In the selector counter circuit **230**D, the counter **235** counts the first comparison result signal Vout**11** from the comparator **220** in the PFM mode. In the selector counter circuit **230**D, the latch **236** latches the second comparison result signal Vout**12** in synchronization with a clock having a predetermined frequency in the dual sampling read-out mode.
(140) The fifth embodiment can produce the same effects as the above-described first to third embodiments.
(141) <Example of Application of Solid-State Imaging Device>
(142) FIG. **20** illustrates an example of a device to which the solid-state imaging devices relating to the embodiments of the present invention are applied. FIG. **21** is a timing chart schematically showing how the applied device shown in FIG. **20** works.
(143) The applied device **300** shown in FIG. **20** includes an image sensor **310**, to which the solid-state imaging device **10** (**10**A to **10**D) relating to any one of the first to fifth embodiments is applied, a light emitter **320** and a control unit **333**.
(144) The applied device **300** is configured to detect whether an object (event) OBJ is present or absent within a designated area by taking advantage of the repeatedly performed operations in the pixel circuits included in the image sensor **310**. Specifically speaking, in the PFM mode, in the exposure period PEXP, overflowing of the charges from the photodiode PD**11** to the floating diffusion FD**11**, self-resetting of the floating diffusion FD**11** resulting from the feedback resetting through the comparison result signal Vout from the comparator **220**, and counting of the number of times the floating diffusion FD**11** is reset (the reset frequency of the floating diffusion FD**11**) by the selector counter circuit **230** are performed repeatedly

(145) A specific example is described. Under a very high illuminance condition, the light emitter **320** emits detection light DL toward a designated region RGN. If an object OBJ is absent within the designated region RGN, counting of the number of times the floating diffusion FD**11** is reset (the reset frequency of the floating diffusion FD**11**) is repeatedly performed at predetermined intervals. If the object OBJ enters the designated region RGN, the object OBJ reflects the detection light DL, and the reflected light is received by the pixel part **20** including the pixel circuits **200**. The intensity of the received light resultantly changes and takes a different value than when the object OBJ is absent, and it takes longer for the potential of the floating diffusion FD**11** to diminish to the reference signal Vout**11**. Stated differently, if the object (event) OBJ is detected, the frequency of the pulse (the number of times) determined by the output from the comparator **220** changes. By detecting whether such a change occurs, the control unit **330** determines whether an event is detected.
(146) <Examples of Application to Electronic Apparatuses>
(147) The solid-state imaging devices **10**, **10**A, **10**B, **10**C and **10**D described above can be applied, as an imaging device, to electronic apparatuses such as digital cameras, video cameras, mobile terminals, surveillance cameras, and medical endoscope cameras.
(148) FIG. **22** shows an example configuration of an electronic apparatus including a camera system to which the solid-state imaging devices according to the embodiments of the present invention can be applied.
(149) As shown in FIG. **22**, an electronic apparatus **400** includes a CMOS image sensor **410** that can be constituted by the solid-state imaging devices **10**, **10**A, **10**B, **10**C and **10**D relating to the embodiments of the present invention. The electronic apparatus **400** further includes an optical system (such as a lens) **420** for redirecting the incident light to the pixel region of the CMOS image sensor **410** (to form a subject image). The electronic apparatus **400** includes a signal processing circuit (PRC) **430** for processing the output signals from the CMOS image sensor **410**.
(150) The signal processing circuit **430** performs predetermined signal processing on the output signals from the CMOS image sensor **410**. The image signals resulting from the processing in the signal processing circuit **430** can be handled in various manners. For example, the image signals can be displayed as a video image on a monitor having a liquid crystal display, printed by a printer, or recorded directly on a storage medium such as a memory card.
(151) As described above, a high-performance, compact, and low-cost camera system can be provided that includes the above-described solid-state imaging device **10**, **10**A, **10**B, **10**C or **10**D as the CMOS image sensor **410**. Accordingly, the embodiments of the present invention can provide for electronic apparatuses such as surveillance cameras and medical endoscope cameras, which are used for applications where the cameras are installed under restricted conditions from various perspectives such as the installation size, the number of connectable cables, the length of cables and the installation height.
LIST OF REFERENCE NUMBERS
(152) **10**, **10**A, **10**B, **10**C, **10**D: solid-state imaging device **20**: pixel part **200**: pixel circuit PD**11**: photodiode FD**11**: floating diffusion TG**11-**Tr: transfer transistor RST**11-**Tr, RST**12-**Tr: reset transistor SF**11-**Tr: source follower transistor SG**11-**Tr: storage transistor CS**11**: storage capacitor AB**11-**Tr: anti-blooming transistor **220**, **220**B, **220**D: comparator **230**, **230**C, **230**D: selector counter circuit **231**: selector **232**: counter **233**: up-down counter **234**: selector **235**: counter **236**: latch **300**: applied device (event detecting device) **310**: image sensor **320**: light emitter **330**: control unit **400**: electronic apparatus **410**: CMOS image sensor **420**: optical system **430**: signal processing circuit (PRC)

Claims

1. A solid-state imaging device comprising: a readable pixel configured to perform photoelectric conversion, the readable pixel being configured to produce a readable signal corresponding to an illuminance condition of incident light; a comparator configured to compare a voltage signal read from the readable pixel against a reference signal and output a comparison result signal corresponding to a result of the comparison; and a selector counter circuit including a selector circuit and a counter circuit, the selector circuit being configured to select an external clock or an output from the comparator, the counter circuit being configured to count an output from the selector circuit, wherein the readable pixel includes: a photoelectric conversion element configured to store therein, in an exposure period, charges corresponding to an amount of the incident light; a transfer element configured to remain in a non-conduction state during the exposure period and to remain in a conduction state during a transfer period to transfer the charges stored in the photoelectric conversion element; a floating diffusion configured to hold the charges transferred thereto via the transfer element so that the charges are read out as a voltage signal; at least one reset element configured to perform a reset operation of discharging the charges stored in the floating diffusion; and an overflow path configured to allow charges overflowing from the photoelectric conversion element to overflow toward a region where the floating diffusion is formed, wherein the readable pixel, the comparator, the selector circuit and the counter circuit: under a high illuminance condition, if the voltage signal corresponding to the charges held in the floating diffusion reaches reference voltage, perform a pulse frequency modulation (PFM) mode, according to which the comparison result signal from the comparator is applied to perform feedback resetting to allow the floating diffusion to perform self-resetting, and the selector counter circuit counts a number of times the floating diffusion is reset (a reset frequency of the floating diffusion) in response to the comparison result signal; and under a low illuminance condition, perform a dual sampling read-out mode using the charges stored in the photoelectric conversion element and overflow charges, wherein the readable pixel and the selector counter circuit are configured to selectively operate in a PFM mode or dual sampling read-out mode according to a mode selection signal, wherein, in the PFM mode, a first reference signal of a fixed level is fed to the comparator, and if the voltage signal corresponding to the charges held in the floating diffusion reaches the level of the first reference signal, the comparator outputs a first comparison result signal to the readable pixel and the selector counter circuit, wherein, in the dual sampling read-out mode, a second reference signal having a ramp waveform showing a continuous change, is fed to the comparator, the comparator compares the voltage signal corresponding to the charges held in the floating diffusion against the second reference signal, and outputs a result of the comparison as a second comparison result signal to the selector counter circuit, and wherein the selector counter circuit: in the PFM mode, counts the first comparison result signal from the comparator; and in the dual sampling read-out mode, latches the second comparison result signal in synchronization with a clock having a predetermined frequency, wherein the readable pixel is configured such that, in the PFM mode, the first comparison result signal is applied as a feedback reset signal to bring the reset element into a conduction state and to reset the floating diffusion.

2. The solid-state imaging device of claim 1, wherein the PFM mode is selected under a high illuminance condition and in an exposure period starting after the floating diffusion and the photoelectric conversion element are reset, and wherein, following the PFM mode, the dual sampling read-out mode is selected.

3. The solid-state imaging device of claim 1, wherein, in the PFM mode, in the exposure period, overflowing of the charges from the photoelectric conversion element to the floating diffusion, self-resetting of the floating diffusion resulting from feedback resetting through the comparison result signal from the comparator, and counting of reset frequency of the floating diffusion by the selector counter circuit are performed repeatedly.

4. The solid-state imaging device of claim 3, wherein an occurrence of an event changes a reset

pulse frequency.
5. A solid-state imaging device comprising: a readable pixel configured to perform photoelectric conversion, the readable pixel being configured to produce a readable signal corresponding to an illuminance condition of incident light; a comparator configured to compare a voltage signal read from the readable pixel against a reference signal and output a comparison result signal corresponding to a result of the comparison; and a selector counter circuit including a selector circuit and a counter circuit, the selector circuit being configured to select an external clock or an output from the comparator, the counter circuit being configured to count an output from the selector circuit, wherein the readable pixel includes: a photoelectric conversion element configured to store therein, in an exposure period, charges corresponding to an amount of the incident light; a transfer element configured to remain in a non-conduction state during the exposure period and to remain in a conduction state during a transfer period to transfer the charges stored in the photoelectric conversion element; a floating diffusion configured to hold the charges transferred thereto via the transfer element so that the charges are read out as a voltage signal; at least one reset element configured to perform a reset operation of discharging the charges stored in the floating diffusion; and an overflow path configured to allow charges overflowing from the photoelectric conversion element to overflow toward a region where the floating diffusion is formed, wherein the readable pixel, the comparator, the selector circuit and the counter circuit: under a high illuminance condition, if the voltage signal corresponding to the charges held in the floating diffusion reaches reference voltage, perform a pulse frequency modulation (PFM) mode, according to which the comparison result signal from the comparator is applied to perform feedback resetting to allow the floating diffusion to perform self-resetting, and the selector counter circuit counts a number of times the floating diffusion is reset (a reset frequency of the floating diffusion) in response to the comparison result signal; and under a low illuminance condition, perform a dual sampling read-out mode using the charges stored in the photoelectric conversion element and overflow charges, wherein the readable pixel includes: a storage element connected to the floating diffusion; and a storage capacitance element configured to store the charges received from the floating diffusion via the storage element, and wherein the storage element is arranged between the floating diffusion and the reset element.
6. The solid-state imaging device of claim 5, wherein, in the PFM mode, during the self-resetting, the reset element and the storage element remain in a conduction state to reset the floating diffusion and the storage capacitance element.
7. The solid-state imaging device of claim 5, comprising a source follower element serving as an output buffer part configured to convert the charges in the floating diffusion into a voltage signal at a level determined by an amount of the charges and outputting the voltage signal, wherein, by selectively connecting together the floating diffusion and the storage capacitance element via the storage element, a capacitance of the floating diffusion is changed between a first capacitance and a second capacitance to change a conversion gain between a first conversion gain corresponding to the first capacitance and a second conversion gain corresponding to the second capacitance.
8. The solid-state imaging device of claim 7, comprising: a pixel part having a plurality of the readable pixels arranged therein; and a reading part configured to read pixel signals from the plurality of readable pixels in the pixel part, wherein the reading part: if the dual sampling read-out mode is selected following the PFM mode, keeps the storage element in a non-conduction state to disconnect the storage capacitance element from the floating diffusion, so that the charges in the floating diffusion are separated from the charges in the storage capacitance element to switch the gain of the floating diffusion to the first conversion gain corresponding to the first capacitance; in a first reset read-out period following the reset operation, performs a first conversion gain reset read-out operation of reading from the source follower element serving as the output buffer part a first read-out reset signal produced through conversion with the first conversion gain corresponding to the first capacitance of the floating diffusion and processes the first read-out reset signal in a

predetermined manner; in a first read-out period following a first transfer period following the first reset read-out period, performs a first conversion gain read-out operation of reading from the source follower element serving as the output buffer part a first read-out signal produced through conversion with the first conversion gain corresponding to the first capacitance of the floating diffusion and processing the first read-out signal in a predetermined manner; after performing the first conversion gain read-out operation, keeps the storage element in a conduction state to connect the storage capacitance element to the floating diffusion, so that the charges in the floating diffusion and the charges in the storage capacitance element are combined together to switch the gain of the floating diffusion to the second conversion gain corresponding to the second capacitance; in a second read-out period following a second transfer period following the first read-out period, performs a second conversion gain read-out operation of reading from the source follower element serving as the output buffer part a second read-out signal produced through conversion with the second conversion gain corresponding to the second capacitance of the floating diffusion and processing the second read-out signal in a predetermined manner; and after the reset element resets the floating diffusion, performs a second conversion gain reset read-out operation of reading from the source follower element serving as the output buffer part a second read-out reset signal produced through conversion with the second conversion gain corresponding to the second capacitance of the floating diffusion serving as an output node and processing the second read-out reset signal in a predetermined manner.

9. The solid-state imaging device of claim 8, wherein the reading part performs the first conversion gain reset read-out operation in the exposure period.

10. The solid-state imaging device of claim 8, wherein the reading part performs the second conversion gain reset read-out operation and the second conversion gain read-out operation under a very low illuminance condition.

11. The solid-state imaging device of claim 8, wherein the reading part performs the first conversion gain reset read-out operation, the first conversion gain read-out operation, the second conversion gain read-out operation and the second conversion gain reset read-out operation under a medium illuminance condition.

12. The solid-state imaging device of claim 7, wherein the source follower element is constituted by a source follower transistor having a gate connected to the floating diffusion and a drain connected to a predetermined potential, and wherein the comparator includes: input terminals one of which is connected to a source of the source follower transistor via a coupling capacitor and the other of which is connected to an input line of the reference signal; and an auto-zero switch connected between an output terminal of the comparator and the input terminal of the comparator connected to the source of the source follower transistor.

13. A solid-state imaging device comprising: a readable pixel configured to perform photoelectric conversion, the readable pixel being configured to produce a readable signal corresponding to an illuminance condition of incident light; a comparator configured to compare a voltage signal read from the readable pixel against a reference signal and output a comparison result signal corresponding to a result of the comparison; and a selector counter circuit including a selector circuit and a counter circuit, the selector circuit being configured to select an external clock or an output from the comparator, the counter circuit being configured to count an output from the selector circuit, wherein the readable pixel includes: a photoelectric conversion element configured to store therein, in an exposure period, charges corresponding to an amount of the incident light; a transfer element configured to remain in a non-conduction state during the exposure period and to remain in a conduction state during a transfer period to transfer the charges stored in the photoelectric conversion element; a floating diffusion configured to hold the charges transferred thereto via the transfer element so that the charges are read out as a voltage signal; at least one reset element configured to perform a reset operation of discharging the charges stored in the floating diffusion; and an overflow path configured to allow charges overflowing from the photoelectric

conversion element to overflow toward a region where the floating diffusion is formed, wherein the readable pixel, the comparator, the selector circuit and the counter circuit: under a high illuminance condition, if the voltage signal corresponding to the charges held in the floating diffusion reaches reference voltage, perform a pulse frequency modulation (PFM) mode, according to which the comparison result signal from the comparator is applied to perform feedback resetting to allow the floating diffusion to perform self-resetting, and the selector counter circuit counts a number of times the floating diffusion is reset (a reset frequency of the floating diffusion) in response to the comparison result signal; and under a low illuminance condition, perform a dual sampling read-out mode using the charges stored in the photoelectric conversion element and overflow charges, wherein the selector counter circuit includes: a counter; and a selector configured to select a signal to be input into the counter in response to a mode selection signal, wherein the selector: when the mode selection signal indicates the PFM mode, feeds the comparison result signal from the comparator to the counter; and when the mode selection signal indicates the dual sampling read-out mode, feeds a clock pulse having a predetermined frequency to the counter, and wherein the counter: in the PFM mode, counts the comparison result signal from the comparator; and in the dual sampling read-out mode, latches a second comparison result signal in synchronization with the clock pulse.

14. A solid-state imaging device comprising: a readable pixel configured to perform photoelectric conversion, the readable pixel being configured to produce a readable signal corresponding to an illuminance condition of incident light; a comparator configured to compare a voltage signal read from the readable pixel against a reference signal and output a comparison result signal corresponding to a result of the comparison; and a selector counter circuit including a selector circuit and a counter circuit, the selector circuit being configured to select an external clock or an output from the comparator, the counter circuit being configured to count an output from the selector circuit, wherein the readable pixel includes: a photoelectric conversion element configured to store therein, in an exposure period, charges corresponding to an amount of the incident light; a transfer element configured to remain in a non-conduction state during the exposure period and to remain in a conduction state during a transfer period to transfer the charges stored in the photoelectric conversion element; a floating diffusion configured to hold the charges transferred thereto via the transfer element so that the charges are read out as a voltage signal; at least one reset element configured to perform a reset operation of discharging the charges stored in the floating diffusion; and an overflow path configured to allow charges overflowing from the photoelectric conversion element to overflow toward a region where the floating diffusion is formed, wherein the readable pixel, the comparator, the selector circuit and the counter circuit: under a high illuminance condition, if the voltage signal corresponding to the charges held in the floating diffusion reaches reference voltage, perform a pulse frequency modulation (PFM) mode, according to which the comparison result signal from the comparator is applied to perform feedback resetting to allow the floating diffusion to perform self-resetting, and the selector counter circuit counts a number of times the floating diffusion is reset (a reset frequency of the floating diffusion) in response to the comparison result signal; and under a low illuminance condition, perform a dual sampling read-out mode using the charges stored in the photoelectric conversion element and overflow charges, wherein the selector counter circuit includes: an up-down counter; and a selector configured to select a signal to be input into a clock terminal of the up-down counter in response to the comparison result signal output from the comparator, wherein the selector has: a first input terminal connected to a reference potential; a second input terminal configured to be set at a fixed potential in the PFM mode and configured to be connected to a feeding line of a signal selected as a clock pulse having a predetermined frequency in the dual sampling read-out mode; and an output terminal connected to a clock terminal of the up-down counter, wherein, if the output from the comparator flips in the PFM mode, the selector disconnects connection between the output terminal and the first input terminal and establishes connection between the output terminal and the second

input terminal, and wherein, if the output from the comparator flips in the dual sampling read-out mode, the selector disconnects connection between the output terminal and the second input terminal and establishes connection between the output terminal and the first input terminal.

15. A solid-state imaging device comprising: a readable pixel configured to perform photoelectric conversion, the readable pixel being configured to produce a readable signal corresponding to an illuminance condition of incident light; a comparator configured to compare a voltage signal read from the readable pixel against a reference signal and output a comparison result signal corresponding to a result of the comparison; and a selector counter circuit including a selector circuit and a counter circuit, the selector circuit being configured to select an external clock or an output from the comparator, the counter circuit being configured to count an output from the selector circuit, wherein the readable pixel includes: a photoelectric conversion element configured to store therein, in an exposure period, charges corresponding to an amount of the incident light; a transfer element configured to remain in a non-conduction state during the exposure period and to remain in a conduction state during a transfer period to transfer the charges stored in the photoelectric conversion element; a floating diffusion configured to hold the charges transferred thereto via the transfer element so that the charges are read out as a voltage signal; at least one reset element configured to perform a reset operation of discharging the charges stored in the floating diffusion; and an overflow path configured to allow charges overflowing from the photoelectric conversion element to overflow toward a region where the floating diffusion is formed, wherein the readable pixel, the comparator, the selector circuit and the counter circuit: under a high illuminance condition, if the voltage signal corresponding to the charges held in the floating diffusion reaches reference voltage, perform a pulse frequency modulation (PFM) mode, according to which the comparison result signal from the comparator is applied to perform feedback resetting to allow the floating diffusion to perform self-resetting, and the selector counter circuit counts a number of times the floating diffusion is reset (a reset frequency of the floating diffusion) in response to the comparison result signal; and under a low illuminance condition, perform a dual sampling read-out mode using the charges stored in the photoelectric conversion element and overflow charges, wherein the selector counter circuit includes: a counter; and a latch, wherein, in the PFM mode, the counter counts a first comparison result signal from the comparator, and wherein, in the dual sampling read-out mode, the latch latches a second comparison result signal in synchronization with a clock having a predetermined frequency.

16. A solid-state imaging device comprising: a readable pixel configured to perform photoelectric conversion, the readable pixel being configured to produce a readable signal corresponding to an illuminance condition of incident light; a comparator configured to compare a voltage signal read from the readable pixel against a reference signal and output a comparison result signal corresponding to a result of the comparison; and a selector counter circuit including a selector circuit and a counter circuit, the selector circuit being configured to select an external clock or an output from the comparator, the counter circuit being configured to count an output from the selector circuit, wherein the readable pixel includes: a photoelectric conversion element configured to store therein, in an exposure period, charges corresponding to an amount of the incident light; a transfer element configured to remain in a non-conduction state during the exposure period and to remain in a conduction state during a transfer period to transfer the charges stored in the photoelectric conversion element; a floating diffusion configured to hold the charges transferred thereto via the transfer element so that the charges are read out as a voltage signal; at least one reset element configured to perform a reset operation of discharging the charges stored in the floating diffusion; and an overflow path configured to allow charges overflowing from the photoelectric conversion element to overflow toward a region where the floating diffusion is formed, wherein the readable pixel, the comparator, the selector circuit and the counter circuit: under a high illuminance condition, if the voltage signal corresponding to the charges held in the floating diffusion reaches reference voltage, perform a pulse frequency modulation (PFM) mode, according to which the

comparison result signal from the comparator is applied to perform feedback resetting to allow the floating diffusion to perform self-resetting, and the selector counter circuit counts a number of times the floating diffusion is reset (a reset frequency of the floating diffusion) in response to the comparison result signal; and under a low illuminance condition, perform a dual sampling read-out mode using the charges stored in the photoelectric conversion element and overflow charges, comprising a gate element configured to transfer the charges stored in the photoelectric conversion element toward a region other than the region where the floating diffusion is formed.

17. A method for driving a solid-state imaging device, the solid-state imaging device including: a readable pixel configured to perform photoelectric conversion, the readable pixel being configured to produce a readable signal corresponding to an illuminance condition of incident light; a comparator configured to compare a voltage signal read from the readable pixel against a reference signal and output a comparison result signal corresponding to a result of the comparison; and a selector counter circuit including a selector circuit and a counter circuit, the selector circuit being configured to select an external clock or an output from the comparator, the counter circuit being configured to count an output from the selector circuit, wherein the readable pixel includes: a photoelectric conversion element configured to store therein, in an exposure period, charges corresponding to an amount of the incident light; a transfer element configured to remain in a non-conduction state during the exposure period and to remain in a conduction state during a transfer period to transfer the charges stored in the photoelectric conversion element; a floating diffusion configured to hold the charges transferred thereto via the transfer element so that the charges are read out as a voltage signal; at least one reset element configured to perform a reset operation of discharging the charges stored in the floating diffusion; and an overflow path configured to allow charges overflowing from the photoelectric conversion element to overflow toward a region where the floating diffusion is formed, wherein, under a high illuminance condition, if the voltage signal corresponding to the charges held in the floating diffusion reaches reference voltage, a pulse frequency modulation (PFM) mode is performed, according to which the comparison result signal from the comparator is applied to perform feedback resetting to allow the floating diffusion to perform self-resetting, and the selector counter circuit counts a reset frequency of the floating diffusion in response to the comparison result signal, and wherein, under a low illuminance condition, a dual sampling read-out mode is performed using the charges stored in the photoelectric conversion element and overflow charges, wherein selectively operate the readable pixel and the selector counter circuit in the PFM mode or the dual sampling read-out mode according to a mode selection signal, wherein, in the PFM mode, feed a first reference signal of a fixed level to the comparator, and if the voltage signal corresponding to the charges held in the floating diffusion reaches the level of the first reference signal, output a first comparison result signal to the readable pixel and the selector counter circuit, wherein, in the dual sampling read-out mode, feed a second reference signal having a ramp waveform showing a continuous change to the comparator, compare the voltage signal corresponding to the charges held in the floating diffusion against the second reference signal, and output a result of the comparison as a second comparison result signal to the selector counter circuit, and wherein, in the PFM mode, count the first comparison result signal from the comparator; and wherein, in the dual sampling read-out mode, latch the second comparison result signal in synchronization with a clock having a predetermined frequency, wherein, in the PFM mode, apply the first comparison result signal as a feedback reset signal to bring the reset element into a conduction state and to reset the floating diffusion.

18. An electronic apparatus comprising: a solid-state imaging device; and an optical system for forming a subject image on the solid-state imaging device, wherein the solid-state imaging device includes: a readable pixel configured to perform photoelectric conversion, the readable pixel being configured to produce a readable signal corresponding to an illuminance condition of incident light; a comparator configured to compare a voltage signal read from the readable pixel against a reference signal and output a comparison result signal corresponding to a result of the comparison;

and a selector counter circuit including a selector circuit and a counter circuit, the selector circuit being configured to select an external clock or an output from the comparator, the counter circuit being configured to count an output from the selector circuit, wherein the readable pixel includes: a photoelectric conversion element configured to store therein, in an exposure period, charges corresponding to an amount of the incident light; a transfer element configured to remain in a non-conduction state during the exposure period and to remain in a conduction state during a transfer period to transfer the charges stored in the photoelectric conversion element; a floating diffusion configured to hold the charges transferred thereto via the transfer element so that the charges are read out as a voltage signal; at least one reset element configured to perform a reset operation of discharging the charges stored in the floating diffusion; and an overflow path configured to allow charges overflowing from the photoelectric conversion element to overflow toward a region where the floating diffusion is formed, wherein the readable pixel, the comparator, the selector circuit and the counter circuit: under a high illuminance condition, if the voltage signal corresponding to the charges held in the floating diffusion reaches reference voltage, perform a pulse frequency modulation (PFM) mode, according to which the comparison result signal from the comparator is applied to perform feedback resetting to allow the floating diffusion to perform self-resetting, and the selector counter circuit counts a reset frequency of the floating diffusion in response to the comparison result signal; and under a low illuminance condition, perform a dual sampling read-out mode using the charges stored in the photoelectric conversion element and overflow charges, wherein the readable pixel and the selector counter circuit are configured to selectively operate in a PFM mode or dual sampling read-out mode according to a mode selection signal, wherein, in the PFM mode, a first reference signal of a fixed level is fed to the comparator, and if the voltage signal corresponding to the charges held in the floating diffusion reaches the level of the first reference signal, the comparator outputs a first comparison result signal to the readable pixel and the selector counter circuit, wherein, in the dual sampling read-out mode, a second reference signal having a ramp waveform showing a continuous change, is fed to the comparator, the comparator compares the voltage signal corresponding to the charges held in the floating diffusion against the second reference signal, and outputs a result of the comparison as a second comparison result signal to the selector counter circuit, and wherein the selector counter circuit: in the PFM mode, counts the first comparison result signal from the comparator; and in the dual sampling read-out mode, latches the second comparison result signal in synchronization with a clock having a predetermined frequency, wherein the readable pixel is configured such that, in the PFM mode, the first comparison result signal is applied as a feedback reset signal to bring the reset element into a conduction state and to reset the floating diffusion.
